TF-201 HEATING JACKET - PRESSURE DROP OPTIMIZATION

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FIXED OPERATING CONDITIONS:

  Pyrolizer diameter:        0.50 m (500 mm)
  Pyrolizer length:          4.20 m
  HTF mass flow rate:        0.25 kg/s (Helisol 5A)
  HTF inlet temperature:     302°C (575.2 K)
  Wall temperature:          280°C (553.2 K)
  Required heat duty:        11.29 kW
  Fanning friction factor:   0.009
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DESIGN COMPARISON

Parameter	Baseline	Initial Best	Extended Best	Units
Jacket Diameter	530	528	552	mm
Channel Width (w)	50	118	62	mm
Rib Thickness (t)	50	100	330	mm
Pitch (w + t)	100	218	392	mm
Number of Turns	42	19	11	-
Annular Gap	15	14	26	mm
Hydraulic Diameter	23	25	37	mm
Fluid Velocity	0.97	0.43	0.22	m/s
Coil Length	54.2	31.0	17.8	m
Pressure Drop	8.56	0.80	0.31	kPa
DP Reduction	-	90.7%	96.4%	-
Design Valid	Yes	Yes	Yes	-

DESIGN DESCRIPTIONS:

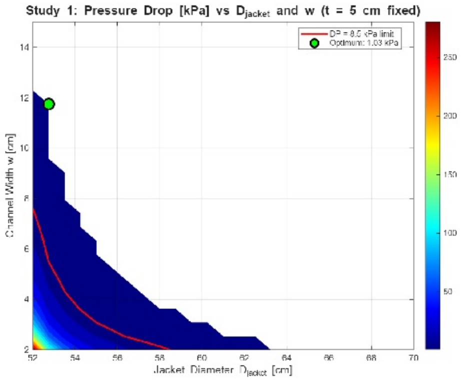
Baseline Design: Original configuration with 5cm channel width and 5cm rib thickness. Pressure drop (8.56 kPa) exceeds the 8.5 kPa design limit. High number of helical turns (42) results in long coil path and high pressure losses.

Initial Best (Study 3): Optimized by increasing channel width to 11.8 cm and rib thickness to 10 cm. Reduces turns to 19 and achieves 90.7% pressure drop reduction. Increased pitch reduces total coil length.

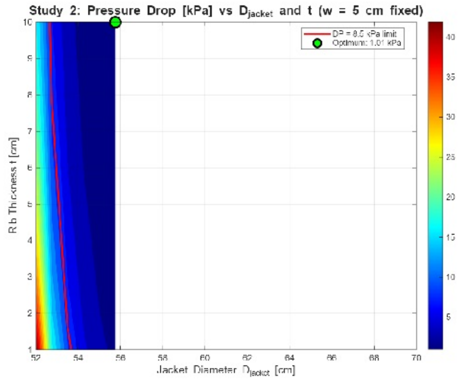
Extended Best: Further optimization with 33 cm rib thickness dramatically reduces the number of turns to only 11. Achieves 96.4% pressure drop reduction to 0.31 kPa. Most aggressive design with lowest pumping requirements.

INITIAL PARAMETRIC STUDY: D_{jacket}, w, t OPTIMIZATION

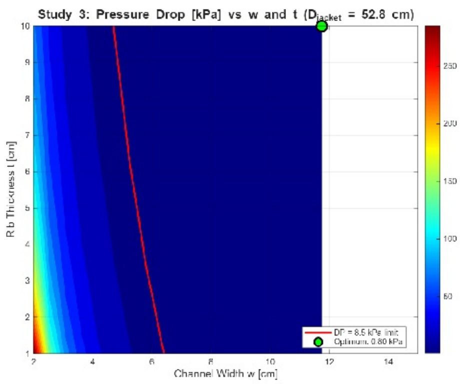
D_{jacket} vs Channel Width



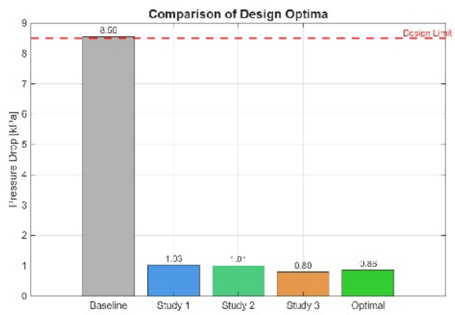
D_{jacket} vs Rib Thickness



Channel Width vs Rib Thickness



Design Comparison



KEY FINDINGS - INITIAL STUDY:

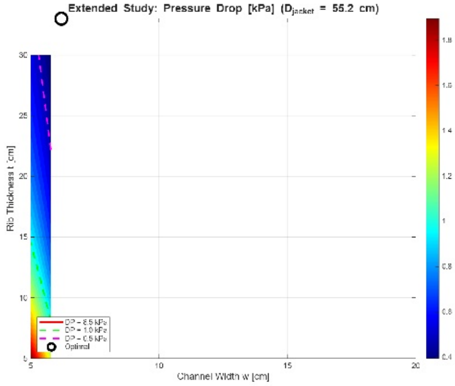
Increasing channel width (w) dramatically reduces pressure drop by increasing flow area and hydraulic diameter

Larger jacket diameter has secondary effect - increases annular gap and flow cross-section

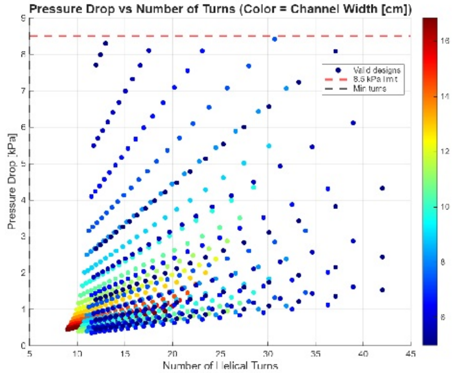
Rib thickness (t) shows potential for further optimization - optimal was at edge of search range (10 cm)

ENHANCED OPTIMIZATION: EXTENDED PARAMETER RANGES

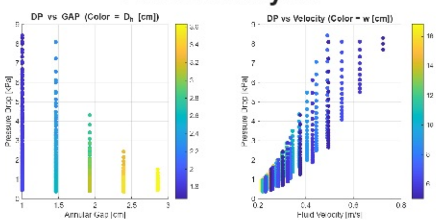
Extended w vs t Search



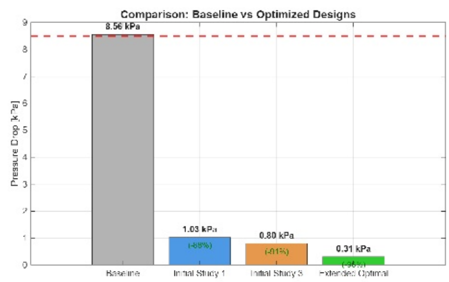
Pressure Drop vs Number of Turns



Pareto Analysis



Final Optimization Comparison



KEY FINDINGS - ENHANCED STUDY:

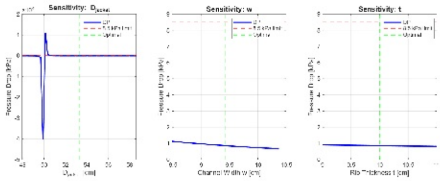
RIB THICKNESS (t) is the dominant parameter - increasing to 33 cm reduces turns from 42 to 11

Fewer helical turns directly reduces total coil length, which scales linearly with pressure drop

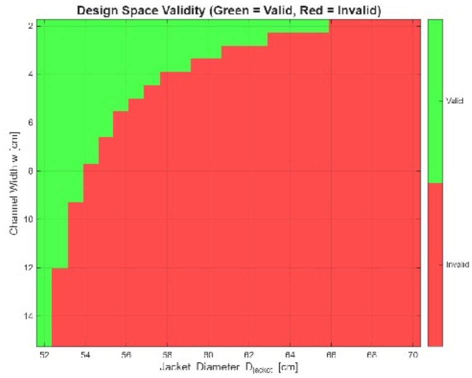
Lower fluid velocity (0.22 m/s vs 0.97 m/s) further reduces DP since it scales with velocity squared

PHYSICAL ANALYSIS: UNDERSTANDING THE OPTIMIZATION

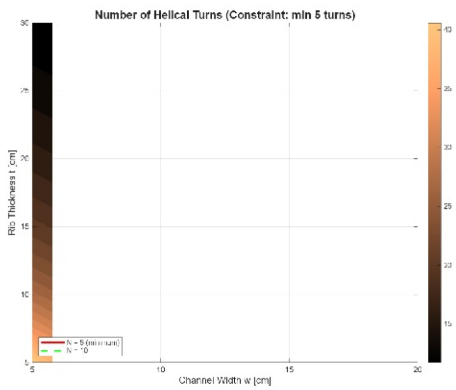
Parameter Sensitivity Analysis



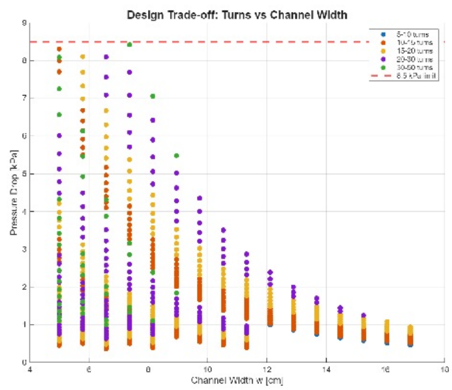
Design Space Validity



Number of Helical Turns



Design Trade-offs



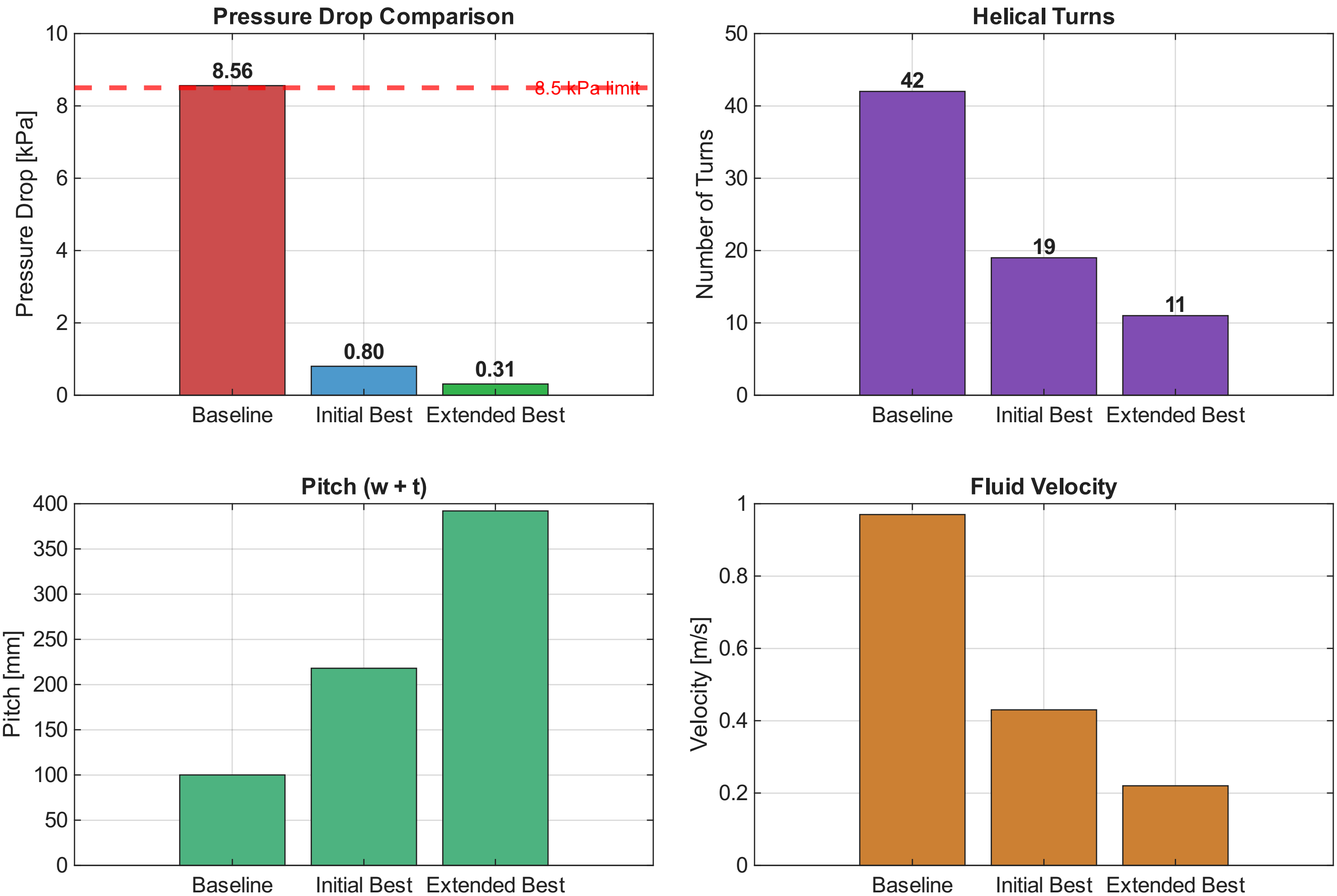
PRESSURE DROP PHYSICS: DP = 4 * f_c * ($L_c$oil / D_h) * (rho * v^2 / 2)

$L_c$oil = $N_t$urns * $L_t$urn : Reducing turns has the strongest effect on DP

v = $m_d$ot / (rho * GAP * w) : Larger flow area reduces velocity, DP scales with v^2

D_h = 2*GAP*w/(GAP+w) : Larger hydraulic diameter reduces friction losses

DESIGN RECOMMENDATIONS



RECOMMENDED DESIGNS:

OPTION A - Aggressive (Extended Best): $D_j$acket=55cm, w=6cm, t=33cm
  Pressure Drop: 0.31 kPa (96% reduction)
  Best for: Maximum pumping efficiency, simplest fabrication (only 11 turns)
  Consideration: Verify heat transfer uniformity with CFD

OPTION B - Balanced: $D_j$acket=55cm, w=8cm, t=25cm
  Pressure Drop: ~0.4 kPa (95% reduction)
  Best for: Balance between DP reduction and manufacturing practicality

OPTION C - Conservative (Initial Best): $D_j$acket=53cm, w=12cm, t=10cm
  Pressure Drop: 0.80 kPa (91% reduction)
  Best for: More conventional geometry, easier to validate

AVOID BASELINE DESIGN:
  Exceeds the 8.5 kPa pressure drop limit with no safety margin.

All optimized designs maintain thermal validity ($A_a$vailable >= $A_r$equired).